DEPARTMENT EXPECTATIONS

Competitive Programming

Led by **Kavish**

Find the pattern, think through the trade-offs, and solve the problem.

CP Induction 2026–27 · Preparation syllabus for first- and second-year students.

First-year students

- Focus on programming fundamentals, logical reasoning, and basic DSA.

Round 1 · Programming basics

- Variables, data types, operators, loops, and conditions
- Functions and basic recursive functions
- Arrays and strings
- Basic number problems: prime, palindrome, factorial, and similar problems

Round 1 · Data structures and algorithms

- Searching: linear search and binary search
- Sorting: bubble sort, selection sort, and insertion sort
- Array problems: reversal, second largest, missing number, and similar problems
- Basic greedy problems

Round 1 · MCQs and pseudocode

- Output prediction and dry runs
- Loop tracing and debugging
- Programming fundamentals and DSA basics
- Basic time-complexity analysis

Round 2 · Technical interview

- Programming fundamentals and basic DSA
- Approach to solving a coding problem
- Code and logic explanation
- Communication, consistency, and eagerness to learn

Second-year students

- Expect stronger DSA proficiency, algorithmic problem-solving, and knowledge of advanced topics, including dynamic programming.

Round 1 · Core DSA

- Arrays, strings, hashing, and prefix sums
- Two pointers and sliding window
- Binary search and sorting
- Linked lists, stacks, and queues
- Trees, binary search trees (BST), and heaps
- Graphs: BFS, DFS, and shortest path
- Greedy algorithms
- Bit manipulation
- Dynamic programming

Round 1 · MCQs and pseudocode

- Time and space complexity
- Code tracing and debugging
- Data structures and algorithms
- DP recurrence and graph traversal logic

Round 2 · Technical interview

- DSA concepts and complexity analysis
- Improving brute-force solutions
- Problem-solving and code implementation
- Explaining approaches and edge cases
- Communication, collaboration, and commitment

Selection criteria

- Candidates are evaluated on technical knowledge, problem-solving ability, code quality, communication skills, and eagerness to learn.

Submitting an application does not confirm selection. Shortlisting and an offline round follow, and their date, time, and venue will be announced.